# Arithmetic determination of relations among lattice theta series

A formal Tate and finite-cover construction with Lean verification

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**Abstract**

We give an effective arithmetic construction of the complete relation kernel, an image basis and exact rational coordinates for any nonempty finite family of positive-definite even integral lattice theta series of a common rank. An integral orthogonal sublattice expresses each series as a finite sum of products of unary residue theta series. A formal product and norm calculation identifies a common power of every unary factor with a section obtained from Kato's algebraic theta unit and its initial term at the zero section. Normalization in a common Kummer extension then produces regular sections of one line bundle, and a divisor-degree argument supplies an explicit finite coefficient bound. No Poisson summation, complex theta transformation, or analytic valence formula enters the construction; the finite coefficient bound is derived directly from algebraic divisor degrees. We also prove exact stopping criteria that allow certified termination before the general bound, and give examples with zero and nonzero theta kernels. An additional torsion-level normalization, using exact projective theta ratios and four-torsion coordinates, replaces the initial $K^s$ cover factor by a common $2^{r-1}$ cover and yields a much smaller explicit coefficient bound.

The constructed sections are modular forms of weight $n/2$ in the standard arithmetic sense on a common finite level of the algebraic metaplectic moduli stack. We construct that level as an actual finite etale moduli cover, identify the tautological half-Hodge line, prove regularity at every cusp, and compute all prescribed Tate expansions. Thus pivot columns give a modular-form basis of the theta image with exact rational coordinates. The construction uses arithmetic moduli and formal completions throughout, allowing a finite metaplectic level that may be noncongruence. A Lean 4.33.1 development verifies the complete chain from lattice decompositions and normalized Tate expansions to the geometric coefficient bounds, certified stopping criteria and exact kernel and image computations.

## 1 Introduction

Given a finite family of positive-definite even integral lattices of the same rank, we seek all complex linear relations among their formal theta series, a basis of the theta image, and exact coordinates in that basis. This paper constructs these outputs through arithmetic geometry and exact finite linear algebra, and supplies a Lean 4.33.1 formalization of the stated proof chain. Both a conservative and a sharper explicit coefficient bound are retained.

The historical motivation is Eichler's ICM 1950 report [1, p. 70], which asks for a "purely arithmetical theory of formal $\theta$-series" and discusses their linear relations and expression in an independent family of modular forms. Motivated by this programme, we study a precise constructive problem: determine the complete relation kernel and image coordinates for an arbitrary finite input family by an explicit arithmetic construction.

### 1.1 Classical modularity and finite determination

Once the theta series are known to lie in a common space of modular forms and an effective vanishing bound $B$ is available, their full relation kernel is the kernel of the finite coefficient matrix $M_B$. Since the matrix has integer entries, exact rational elimination gives a rational basis whose complex span is the full kernel, independent pivot theta series spanning the image, and unique rational coordinates for every input series. This deduction is already part of the classical computational framework. Sturm's congruence theorem [2] and its computational treatment in [3, Section 9.4.1.2, Theorem 9.18] are central precedents for finite coefficient tests. This approach requires compatible common levels and characters, together with a suitable vanishing theorem or weight reduction in half-integral weight. Sturm-type bounds also admit algebraic treatments. Here we derive the finite bound directly from the degrees of explicitly constructed line bundles and identify their sections with the prescribed formal theta series.

There are also direct results on theta relations. Choie and Chung [4] compute dimensions, independent families and relations for genus-averaged ternary theta series at levels $4D$ with $D$ squarefree. Zhu [5, Theorem 7.10] gives a finite-vector criterion for all linear relations among specified powers of Jacobi theta series with characteristics attached to an even lattice, together with computational implementations. Its algorithmic criterion uses modularity and a valence bound. Saha and Hanke [6, Theorem 1.1] classify three-term relations for positive-definite binary quadratic forms. Sugiyama [7] determines bases and dimensions for theta spaces arising from a particular prime-level quaternionic setting using arithmetic geometry. These works provide relation algorithms and basis results for their respective input families and structures. The present construction treats arbitrary finite families of positive-definite even integral Gram matrices of a common rank, with explicit arithmetic normalization and a formally verified connection to the complete kernel and image. The image basis spans precisely the space generated by the supplied theta series.

### 1.2 Existing algebraic theta constructions

Mumford's algebraic theta theory [8] supplies theta groups and their representation-theoretic geometry. His construction of degenerating abelian varieties [13], together with the explicit arithmetic Tate models used here [11], supplies the geometric foundation for our formal product comparisons and global section constructions.

Ramsey [9] develops geometric half-integral-weight modular forms using modular curves and cusp divisors, with $q$-expansions and Hecke operators, starting from the classical theta transformation law. Candelori constructs the theta-multiplier line and its canonical relation to determinant lines [16], and develops theta descent, metaplectic moduli and algebraic half-forms [15]. His algebraic transformation laws and key line-bundle isomorphisms provide the framework for our torsion-level refinement. Our construction fixes the formal frames and scalar normalizations needed for exact unary theta expansions, and verifies regularity at every cusp on the common compact cover. This connects the algebraic theta framework to explicit coefficient bounds for the given lattice family.

### 1.3 The explicit construction and its contribution

We give an explicit arithmetic realization of the complete finite-input problem, tracking the normalizations and geometric degrees needed for effective bounds. The orthogonal coset decomposition of Section 3 reduces the given lattices to a finite collection of unary factors. Section 4 identifies their powers with the complete formal products of Kato's algebraic theta unit [12, 14] and its zero-section initial term. The subsequent isogeny pullback includes the

factor $m^{-K/2}$ forced by the invariant differential. This gives actual global sections of $\omega^{K/2}$ with full expansions exactly equal to $U^K_{m,\nu}$ in the specified Tate frame. Simultaneous roots then place the full lattice theta series in one line bundle on a common finite cover, with regularity checked at every cusp.

The first construction yields

$$B_* = \frac{nK^s I(L)}{48}.$$

The torsion-level refinement in Section 9 fixes the projective theta ratios in a common formal frame, normalizes the key isomorphism, and uses four-torsion coordinates to obtain a common half-Hodge line on a cover of degree at most $2^{r-1}$. It yields

$$B_\dagger = \left\lfloor \frac{n2^{r-1} I(L)}{48L} \right\rfloor.$$

Here $N, L, K, s, r$ and $I(L)$ are defined in Section 2; in particular $s$ counts unary pairs whereas $r$ counts distinct diagonal norms. The refinement replaces the $K^s$ cover-degree estimate of the first construction by $2^{r-1}$ and yields the smaller cutoff $B_\dagger$. Exact early stopping certificates (Lemma 7.1) further permit termination at a shorter prefix whenever full column rank or sufficiently many independent full-series relations have been established.

Together, the normalization bridges, common covers and degree estimates realize the input theta series as prescribed expansions of global sections on one finite metaplectic level. Exact enumeration and rational elimination then determine the full relation kernel, a basis of the theta image and unique rational coordinates. The construction permits noncongruence finite levels; its precise modular-form scope is described in Section 10. For an application to an order, the complete representative list is supplied as input; the determinant-80 example also proves that classification.

### 1.4 The complete formalized proof chain

Lau, Lee and Ono [10] formalize $q$-series results in Lean, including the Jacobi triple product and Rogers–Ramanujan identities. Our development verifies the following connected chain, detailed in Section 11: finite lattice shells and orthogonal cosets; complete formal theta products and normalized algebraic theta units; actual compact covers, common line bundles and all-cusp regularity; divisor-degree coefficient bounds; and exact enumeration and elimination producing the complete complex kernel, a theta-image basis and rational coordinates. The verified assembly includes rank zero, deficient prefixes with sufficient full-series certificates, and the fixed geometric family used at every certified conservative cutoff.

Among the works compared here, we have not identified an earlier formalization covering this complete proof chain for arbitrary nonempty finite families of positive-definite even integral Gram matrices of a common rank. Our Lean development connects the arithmetic constructions and their exact Tate expansions to explicit finite coefficient bounds, certified stopping criteria, and the computation of the full complex relation kernel, a basis of the theta image, and unique rational coordinates. The accompanying declaration register and source-to-proof mapping document this coverage, while joint verification checks the assembled declarations and their dependencies.

## 2 Input and the finite-coefficient theorem

Let $h \geq 1$ and let $A_1, \ldots, A_h$ be symmetric positive-definite integral $n \times n$ matrices with even diagonal and $n > 0$, and put

$$Q_i(x) = \tfrac{1}{2} x^t A_i x, \qquad \theta_i(q) = \sum_{x \in \mathbb{Z}^n} q^{Q_i(x)}.$$

Positive definiteness makes these series well-defined in $\mathbb{Z}[[q]]$. For $n = 0$ all input theta series are 1, so the kernel is simply $\{c : \sum_i c_i = 0\}$ and the image basis is $\{1\}$. The requested order problem is a special case: take the supplied complete list of normalized representatives. The argument does not need an unjustified implication from local elementary divisors to global class completeness. Where representatives are supplied, their completeness remains the stated input hypothesis.

For each $i$, the construction in Section 3 gives an integral matrix $B_i$ of nonzero determinant such that $B_i^t A_i B_i = \mathrm{diag}(m_{i1}, \dots, m_{in})$, with positive even $m_{ij}$. Let $\mathcal{U}$ be the finite collection of distinct unary pairs $(m, \nu)$ occurring in the resulting coset decompositions, with $\nu$ taken modulo $m$. Set

$$N = \mathrm{lcm}_{i,j}\, m_{ij}, \qquad L = 24N^2, \qquad K = (1 + 12N)^2 - 1,$$
$$s = |\mathcal{U}|, \qquad I(L) = L^3 \prod_{p \mid L} (1 - p^{-2}), \qquad B_* = \frac{nK^s I(L)}{48}.$$

The integer $B_*$ is deliberately conservative. Theorem 9.5 proves the substantially smaller bound $B_\dagger = \lfloor n 2^{r-1} I(L)/(48L) \rfloor$, where $r$ counts the distinct diagonal norms $m_{ij}$. The revised implementation uses this latter bound; the first construction is retained as an independent arithmetic route and for comparison.

**Theorem 2.1** (Arithmetic finite-coefficient determination)**.** *Using the algebraic Tate uniformization and algebraic theta-unit theorems specified in Section 4, the construction below gives*

$$\ker_{\mathbb{C}}\big(c \longmapsto \textstyle\sum_i c_i \theta_i\big) = \ker_{\mathbb{C}} M_{B_*}, \qquad M_B = \big([q^a]\theta_i\big)_{0 \le a \le B,\, 1 \le i \le h}.$$

*In particular the kernel is the complex span of an effectively computable rational basis. Pivot columns give a basis of the formal theta image and exact rational coordinates for every theta series in that basis. The same series are the exact Tate expansions of standard arithmetic modular forms of weight $n/2$ on one common finite metaplectic level, regular at every cusp. Consequently the pivot theta series form a linearly independent basis of modular forms for the image. The actual level and its half-Hodge line are established in Corollary 9.6.*

We use the standard arithmetic definition of half-integral-weight modular forms. On a finite representable etale level of the algebraic metaplectic moduli problem, such a form is a section of a tensor power of its tautological half-Hodge line, regular also at the cusps of the compactification. The metaplectic object and its finite-level meaning are specified in [15, Definition 3.1.1 and Remark 3.1.2]. We give the test-object description and verify the finite-level map in Corollary 9.6, so the word "modular" does not mean an arbitrary line bundle endowed with a formal expansion. This permits the usual noncongruence finite-level scope; it does not claim a specified Shimura congruence level or multiplier. All proofs below take place in arithmetic moduli and formal completions. The relation to analytic terminology is discussed separately in Section 10 and is not used in the construction.

## 3 An orthogonal sublattice and unary factors

**Lemma 3.1.** *For an even integral positive-definite matrix $A$, there is an integral $B$ with nonzero determinant and $B^t A B = \mathrm{diag}(m_1, \dots, m_n)$, where each $m_j$ is positive and even. For $H = B^{-1}\mathbb{Z}^n/\mathbb{Z}^n$, choose representatives $t$. Then $\nu_j = m_j t_j$ is integral, and*

$$\theta_A(q) = \sum_{t \in H} \prod_{j=1}^{n} U_{m_j, \nu_j}(q), \qquad U_{m,\nu}(q) = \sum_{r \equiv \nu\,(m)} q^{r^2/(2m)}.$$

*All terms of each product have integral total exponent.*

*Proof.* Rational Gram–Schmidt produces an orthogonal basis over $\mathbb{Q}$. Clearing denominators of its columns produces $B$. Its diagonal norms are even because its columns are integral vectors in the even lattice. The index of $B\mathbb{Z}^n$ in $\mathbb{Z}^n$ is $|\det B|$.

If $x = Bt \in \mathbb{Z}^n$, then $\operatorname{diag}(m_j)t = B^tAx \in \mathbb{Z}^n$. Hence $m_jt_j \in \mathbb{Z}$. The coset of $x$ consists precisely of $B(t+k)$ for $k \in \mathbb{Z}^n$, and

$$Q(B(t+k)) = \sum_j \frac{m_j}{2}(t_j + k_j)^2.$$

Putting $r_j = m_j(t_j + k_j)$ gives the asserted product. Since $B(t+k)$ is integral, its total exponent is integral. Each bounded exponent involves only finitely many vectors, so all regroupings are formal and legitimate. □

For example, for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 12 \end{smallmatrix}\right)$, take $B = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 2 \end{smallmatrix}\right)$. Then $B^tAB = \operatorname{diag}(2, 46)$ and

$$\theta_{x^2+xy+6y^2} = U_{2,0}U_{46,0} + U_{2,1}U_{46,23}.$$

This reduction works without a parity restriction on the rank.

# 4 Formal products and algebraic theta units

We use Kato's algebraic existence and norm characterization [12, Proposition 1.3(1)]: for $c > 1$ prime to 6 there is a base-change compatible rational function ${}_c\vartheta_E$ with divisor $c^2[0] - [E[c]]$, invariant under the norm for multiplication by every $d$ prime to $c$. Its algebraic construction uses divisors and Picard functoriality; see also the modern statement in [14, Theorem 4.1.1]. Kato's original complex proof of the product formula is *not* used here.

For the formal quotient we use the actual positive Tate datum $R = k[[Q]]$, with $k$ of characteristic zero, the torus $\mathbb{G}_m$, and the period lattice $Q^{\mathbb{Z}}$. Its polarization is the symmetric pairing $(Q^r, Q^s) \mapsto Q^{rs}$; for $r \neq 0$, $Q^{r^2}$ lies in the ideal of definition. Thus the completeness, normality and positive polarization hypotheses of Mumford's construction [13, Theorem 3.10, Corollary 4.9, Theorem 4.10] are satisfied. We use the explicit annular model and proper algebraization of [11, Corollary 2.2.4 and Section 2.5]. Its generic fiber is a geometrically connected elliptic curve, with torsion and multiplication represented by $z^d = Q^i$ and $z \mapsto z^d$. No assertion about an unspecified period datum over an arbitrary discrete valuation ring is needed.

**Lemma 4.1** (Constant comparison on a proper formal model)**.** *Let $R = k[[t]]$ have characteristic zero and let $\mathcal{E}/R$ be a proper semistable model with geometrically connected smooth generic fiber. Choose a finite affine formal covering $\widehat{\mathcal{E}} = \bigcup_\alpha \operatorname{Spf}(A_\alpha)$ by noetherian normal domains $A_\alpha$ that are $t$-torsion-free. Suppose compatible elements $u_\alpha \in \operatorname{Frac}(A_\alpha[1/t])$ have no pole at any height-one prime of $A_\alpha[1/t]$. Then they are the restrictions of a constant in $k((t))$.*

*Proof.* The height-one intersection criterion for a noetherian normal domain implies $u_\alpha \in A_\alpha[1/t]$. Choose $b_\alpha \geq 0$ with $t^{b_\alpha}u_\alpha \in A_\alpha$. The covering is finite, so a single $b = \max_\alpha b_\alpha$ works on all charts. Compatibility then gives a global regular function on $\widehat{\mathcal{E}}$. The theorem on formal functions identifies this with an element of $H^0(\mathcal{E}, \mathcal{O}_\mathcal{E})$: this module is finite over the complete ring $R$, hence is already complete. Its restriction to the proper geometrically connected smooth generic fiber is constant. Dividing by $t^b$ proves the assertion. □

Define

$$\gamma_Q(z) = \prod_{r\geq 0}(1 - Q^r z)\prod_{r\geq 1}(1 - Q^r z^{-1}), \qquad \gamma_Q(Qz) = -z^{-1}\gamma_Q(z).$$

For $K_c = c^2 - 1$ and $p_c = -c(c-1)/2$, put

$$f_c(z) = Q^{K_c/12}(-z)^{p_c}\frac{\gamma_Q(z)^{c^2}}{\gamma_Q(z^c)}.$$

We explain why this lemma applies to the comparison used below. For the fixed integer $c > 1$, after a finite scalar extension containing $\mu_c$ and the ramified extension $Q = t^r$, $r = 2c$, use the actual $r$-gon Tate model of [11, Section 2.5]. Its finitely many annular charts have rings

$$A_j = k[[t]]\langle x_j, y_j\rangle/(x_j y_j - t), \qquad 0 \le j < r.$$

On the infinite chain before taking the period quotient, the multiplicative coordinate is $z = t^j x_j$, so $z^{-1} = t^{-j-1}y_j$. The quotient identifies the last overlap by $z \mapsto Qz$. These rings are normal; if necessary refine the cover by completions of algebraic affine charts of the same proper model.

On the $j$th chart the two products in $\gamma_Q(z)$ are

$$\prod_{\ell\ge 0}(1 - t^{r\ell+j}x_j), \qquad \prod_{\ell\ge 1}(1 - t^{r\ell-j-1}y_j).$$

Every exponent is nonnegative. Apart from $1 - x_j$ when $j = 0$ and $1 - y_j$ when $j = r - 1$, all factors form a unit of $A_j$. For $\gamma_Q(z^c)$ the corresponding exponents are $r\ell + cj$ and $E = r\ell - c(j+1)$. Only finitely many $E$ are nonpositive. Since $c \mid r$, a negative $E$ satisfies $E \le -c$, and

$$1 - t^E y_j^c = -t^E y_j^c(1 - t^{-E-c}x_j^c).$$

The monomial is a unit after inverting $t$; the last factor is a unit of $A_j$ unless $E = -c$, when it is $1 - x_j^c$. The case $E = 0$ gives $1 - y_j^c$. All infinite remainders are units. Consequently $\gamma_Q(z^c) \in A_j[1/t]$, and $f_c$ lies in its fraction field. Its only horizontal divisor factors are the simple linear factors of $1 - x_j^c$ or $1 - y_j^c$ and the indicated factors of $\gamma_Q(z)$.

Indeed the $c$-torsion representatives are $z = \zeta_c^b t^{ra/c}$, $0 \le a, b < c$. Since $c \mid r$, these extend to smooth sections of the $r$-gon; the factors just listed are their local equations. Hence the horizontal divisor of $f_c$ is exactly $c^2[0] - [E_Q[c]]$. Kato's algebraic function has the same divisor. On the same normal algebraic model its other divisor components are vertical. After passing to the compatible formal charts and inverting $t$, no vertical component remains; units remain units, and the split smooth torsion sections retain their displayed local equations. Thus the ratio of Kato's function and $f_c$ has order zero at every height-one prime of every $A_j[1/t]$.

Both functions are invariant under $z \mapsto Qz$, so their local ratios agree under the actual quotient overlap maps. Lemma 4.1 therefore makes their ratio a constant over the finite extended coefficient field. The norm computations for $d = 2, 3$ force this constant to satisfy $u^3 = u^8 = 1$, hence $u = 1$. The resulting identity holds already over the original coefficient field. The auxiliary extension $Q = t^{2c}$ is only used in this comparison proof and does not enlarge the later modular level or its degree bound.

**Proposition 4.2** (A formal norm calculation)**.** *If $c > 1$ and $(c, 6) = 1$, then $f_c$ is the full Tate expansion of ${}_c\vartheta_{E_Q}$.*

*Proof.* The translation identity for $\gamma$ proves periodicity of $f_c$. Its divisor is $c^2[0] - [E[c]]$. Let $(d, c) = 1$. The $d^2$ preimages of $z$ are $z_{ij} = \zeta_d^j Q^{i/d} z^{1/d}$ for $0 \le i, j < d$. The finite identity $\prod_j(1 - \zeta_d^j w) = 1 - w^d$ yields

$$\prod_{i,j}\gamma_Q(z_{ij}) = \gamma_Q(z).$$

Write $ci = dk_i + r_i$ with $0 \le r_i < d$, and set

$$S = \sum_i k_i = \frac{(c-1)(d-1)}{2},$$
$$E = \sum_i \left( r_i k_i + \frac{d}{2} k_i(k_i - 1) \right) = \frac{(c^2-1)(d-1)(2d-1)}{12} - \frac{d(c-1)(d-1)}{4}.$$

The latter equality follows by expanding $r_i = ci - dk_i$ and using that the $r_i$ permute $0, \dots, d-1$. Translation gives

$$\prod_{i,j} \gamma_Q(z_{ij}^c) = (-1)^{-S} z^{-cS} Q^{-E} \gamma_Q(z^c).$$

Also $\prod_{i,j}(-z_{ij}) = (-1)^{d^2} z^d Q^{d(d-1)/2}$. The norm ratio therefore has possible factors with exponents

$$(d^2-1)p_c + S, \quad (d-1)p_c + cS, \quad (d^2-1)K_c/12 + p_c d(d-1)/2 + E.$$

The last two are zero. The first is $(c-1)(d-1)(1-c(d+1))/2$, an even integer: check $d$ odd and even separately. Thus $\mathrm{Norm}_{[d]} f_c = f_c$.

The ratio of Kato's function to $f_c$ has no divisor and is a constant $u$ by Lemma 4.1. Norm invariance gives $u^{d^2} = u$. Taking $d = 2, 3$ gives $u^3 = u^8 = 1$, whence $u = 1$. □

Let $B_2(T) = T^2 - T + 1/6$ and define the formal Siegel product

$$g_{a,b}(Q) = -Q^{B_2(a)/2} e^{\pi i b(a-1)} \gamma_Q(Q^a e^{2\pi i b}), \quad 0 \le a < 1.$$

The roots of unity in this notation are algebraic constants. Put $\eta(Q) = Q^{1/24} \prod_{r \ge 1}(1 - Q^r)$. The formal Jacobi triple product gives, for $-m/2 \le \nu < m/2$,

$$U_{m,\nu}(q) = -e^{-\pi i(a-1)/2} \eta(q^m) g_{a,1/2}(q^m), \qquad a = \nu/m + 1/2.$$

The leading exponent is fixed by $1/24 + B_2(a)/2 = (a - 1/2)^2/2$; there is no unspecified formal unit. For completeness, an arithmetic proof of the triple product is as follows. Write $(x;t)_r = \prod_{0 \le j < r}(1 - xt^j)$. Euler coefficient recursions give

$$(-z;t)_\infty = \sum_{r \ge 0} \frac{t^{r(r-1)/2} z^r}{(t;t)_r}.$$

The product $F(z) = (-z;t)_\infty (-t/z;t)_\infty (t;t)_\infty$ satisfies $F(tz) = z^{-1}F(z)$. If $F = \sum a_r z^r$, then $a_r = a_0 t^{r(r-1)/2}$. Its constant coefficient is $(t;t)_\infty \sum_{r \ge 0} t^{r^2}/(t;t)_r^2 = 1$: the sum is the partition generating function, decomposed by the size of its Durfee square. This proves the product identity coefficientwise, with normalization $a_0 = 1$. Substitution $t = Q, z = Q^a$ gives the stated unary identity; bounded exponents still involve finitely many terms, including when $a = 0$.

Take $c = 1 + 12N$ and $K = c^2 - 1$. Evaluating Proposition 4.2 at $z = Q^a e^{2\pi i b}$, with $Na, Nb \in \mathbb{Z}$ and $(a,b) \notin \mathbb{Z}^2$, gives ${}_c\vartheta(P_{a,b}) = g_{a,b}^K$. Indeed the difference of phase exponents is

$$(c-1)(a - c/2) + (c-1)b[1 - (c-1)a] = 12A - 6Nc + 12B - 144AB \in 2\mathbb{Z},$$

where $a = A/N, b = B/N$. Since $24N \mid K$, the remaining triple-product phase also vanishes. Consequently

$$U_{m,\nu}(q)^K = \mathcal{D}_2(q^m)^{K/48} {}_c\vartheta_{E_{q^m}}(P_{\nu/m+1/2,\,1/2}). \tag{1}$$

Here $\mathcal{D}_2(Q) = Q^2 \prod_{r \ge 1}(1 - Q^r)^{48}$. The exponent $K/48$ is integral: $K = 24N(1 + 6N)$ and $N$ is even, because it is the least common multiple of a nonempty family of positive even integers $m_{ij}$. The torsion points actually used above have $b = 1/2$ and are therefore nonzero. Thus (1) is a purely formal product identity so far.

**Lemma 4.3** (An algebraic discriminant square from a theta unit)**.** *On the moduli of elliptic curves in characteristic zero there is a canonical nowhere-vanishing section $D_2$ of $\omega^{24}$ with exact Tate expansion $\mathcal{D}_2(Q)\,(dz/z)^{24}$.*

*Proof.* For $c = 5$, Kato's function has a zero of order 24 along the identity section and no other zero or pole in a neighborhood of it. If $\mathcal{I}$ is the ideal of the identity, then $\mathcal{I}^{24}/\mathcal{I}^{25} \simeq \omega^{24}$. Define $D_2 = 5\,\mathrm{in}_0({}_5\vartheta)$ under this canonical identification. The exact divisor makes this initial term everywhere nonzero, and base-change compatibility makes it canonical.

Put $z = 1+t$ and $P = \prod_{r\geq 1}(1-Q^r)$. Then $\gamma_Q(1+t) = -tP^2 + O(t^2)$ and $\gamma_Q((1+t)^5) = -5tP^2 + O(t^2)$. Proposition 4.2, for which $K_5 = 24$ and $p_5 = -10$, gives

$$ {}_5\vartheta(1+t) = \tfrac{1}{5}Q^2P^{48}t^{24} + O(t^{25}). $$

The class of $t$ is $dz/z$ at the identity. Multiplying the initial term by 5 proves the full expansion, including its constant factor. □

# 5 Algebraic sections and every cusp

Work over a number field containing the required roots of unity and take a geometric component $X$ of the full-level modular curve $X(L)$, $L = 24N^2$, containing the designated Tate cusp. The ambient fine moduli scheme is smooth and projective in characteristic zero: see [11, Theorems 4.1.1 and 4.2.1(2), Lemma 4.2.4]. The selected geometric component and cusp descend to a finite number field extension. Its completed cusp rings have $q = t^L$ [11, Lemma 4.1.7 and (4.2.1)]. The Hodge line on the compactification is the invariant cotangent line at the identity of the smooth locus of the universal generalized elliptic curve. Its cusp frame is $d\log z$. Its universal source elliptic curve has a designated cyclic isogeny $\phi_m : E \to E_m$ of degree $m$, with target $N$-torsion sections. At the designated cusp this is $E_q \to E_{q^m}$, $z \mapsto z^m$ [11, Example 2.5.1]. The level suffices because $mN \mid L$. More explicitly, a target point $P = \zeta_N^B q^{mA/N}$, with integer coordinates $A, B$, has source lift $\widetilde{P} = \zeta_{mN}^B q^{A/N}$, an $mN$-torsion point. These integers are distinct from the rational characteristics $a, b$ used in the formal product identities. The chosen full-level structure supplies this lift and the kernel $\mu_m$ of the quotient isogeny; consequently the selected target $N$-torsion sections exist on this algebraic modular curve.

Let $\omega$ be the Hodge line of the source. In (1), replace the formal $\mathcal{D}_2$ by the algebraic $D_2$ of Lemma 4.3, pull back the target section and multiply by $m^{-K/2}$. This normalization is essential: $\phi_m^*(dz'/z') = m\,dz/z$ at the designated cusp. It defines a section $F_{m,\nu}$ of $\omega^{K/2}$ whose full expansion in the source Tate differential is exactly $U^K_{m,\nu}$.

For clarity, the calculation at each cusp is an independent base change of the same global algebraic theta unit. If $Q_x$ is the target Tate period at a cusp $x$, Proposition 4.2 is applied there with its formal variable $Q$ replaced by $Q_x$. Base-change naturality of Kato's function and of the initial-term construction of $D_2$ puts both factors of the target section in this same completed field. No formula identifying the completed parameters at two different cusps is used.

At another cusp the same nonzero torsion point has Tate coordinates $(a', b')$ with $0 \leq a' < 1$. If $a' = 0$, then $b' \notin \mathbb{Z}$. Thus the leading coefficient of its Siegel product is nonzero, and the order of the target section is

$$ K/24 + KB_2(a')/2 = K(a' - 1/2)^2/2 \geq 0. $$

To justify the local Hodge comparison, write the intersection of the cyclic kernel with the torus as $\mu_e$, $e \mid m$. Quotienting by $\mu_e$ gives $z \mapsto z^e$; the remaining component quotient

enlarges the period lattice and is the identity on torus coordinates. These are actual formal Tate maps [11, Section 2.5], after a finite parameter extension if necessary. Their differential factor is the nonzero constant $e$ in characteristic zero. Pullback multiplies valuations by positive ramification indices and does not add a pole from the Hodge comparison. Hence $F_{m,\nu}$ is regular at every cusp. It is nonzero and nowhere vanishing on the open modular curve: $D_2$ is a unit there, and the nonzero $N$-torsion point is not $c$-torsion.

For the full level problem with a fixed Weil pairing, finite level-structure counting gives at most $I(L)/(2L)$ cusps, each of width $L$. Primitive pairs modulo $L$, up to simultaneous sign, give this count; equivalently divide the $I(L)$ symplectic markings by the cusp stabilizer of order $2L$. The cusps of the selected component form a subset of this finite set. Let $C$ denote their reduced divisor. Naturality of $D_2$ and its exact Tate product prove

$$\operatorname{div}(D_2) = 2LC, \qquad 24 \deg \omega = 2L \deg C \leq I(L).$$

Only this upper bound is needed. We do not consume the proof of geometric connectedness of the entire fixed-pairing modular curve in [11, Theorem 4.2.1(1)], which uses complex connectedness.

## 6 A common root line and an effective jet bound

Set $D = (L/12)C$ and $M = \omega \otimes \mathcal{O}_X(-D)$. Since $24 \mid L$, $D/2$ is an integral divisor. Write $1_D$ for the canonical rational section of $\mathcal{O}_X(D)$, whose divisor is $D$. Then

$$d_M = D_2/1_D^{24} \in H^0(X, M^{24})$$

is nowhere zero. The line-bundle root torsor

$$T = \{v \in M : v^{24} = d_M\} \longrightarrow X$$

is finite etale of degree 24: locally its equation is $z^{24} = a$ with $a$ a unit, and its derivative $24z^{23}$ is invertible. On $T$, put $\varepsilon = v\, 1_D$. It has divisor the pullback of $D$, so it supplies an isomorphism

$$H_T = \mathcal{O}_T(p^*D/2), \qquad H_T^2 \simeq p^*\omega, \qquad 1_{H_T}^2 \longmapsto \varepsilon.$$

The equality of the two divisors makes this a regular isomorphism at every point, not just a rational comparison.

Here the common Tate frame can be specified explicitly. Write $q = t^L$, $E(t) = \prod_{r\geq 1}(1-t^{Lr})$ and $\delta = d\log z$. The regular frames $e_D = t^{-L/12}1_D$ and $e_M = \delta/e_D$ satisfy $d_M = E(t)^{48}e_M^{24}$. Choose the actual formal root $v = E(t)^2 e_M$ and the geometric component $X_0$ containing this branch. The map $p_0 : X_0 \to X$ has degree at most 24 and is unramified at the selected cusp. For $H = H_T|_{X_0}$ the frame

$$h_T = E(t)^{-1}t^{-L/24}1_H$$

satisfies $h_T^2 = \delta$. This construction avoids any parity assumption on the degree of an individual component's Hodge line, and uses no analytic connectedness theorem or eta transformation law.

**Lemma 6.1** (Simultaneous global roots)**.** *Let $X$ be a smooth projective curve in characteristic zero, $H$ a line bundle and $F_1, \dots, F_s$ nonzero sections of $H^K$. Specify roots of these sections in one completed algebraic closure at a point, using the same local frame. There exists a finite map $\pi : Y \to X$ of smooth projective curves of degree at most $K^s$, on which these roots are regular sections of $\pi^*H$ with the specified expansions.*

*Proof.* Choose a rational section $h$ of $H$ and write $F_j = f_j h^K$. In the specified completed algebraic closure choose roots $r_j^K = f_j$ and normalize $X$ in the field generated by all $r_j$. This is one field component, not an arbitrary collection of independently chosen components. We perform this construction over the algebraic closure of the constant field and keep the component containing the prescribed Tate embedding. Thus $Y$ is geometrically connected. Its degree is at most $K^s$; all finitely many chosen algebraic data can subsequently descend to a finite constant extension. The rational sections $s_j = r_j \pi^* h$ satisfy $s_j^K = \pi^* F_j$. At every point $y$, $K \operatorname{ord}_y(s_j) = \operatorname{ord}_y(\pi^* F_j) \geq 0$, proving regularity. The completed embedding gives the requested expansions. □

Apply Lemma 6.1 over $X_0$ to the pulled-back sections $p_0^* F_j$ and the just constructed common frame $h_T$. Prescribe $r_j = U_j(h_T/h)$ simultaneously. All these roots belong to the same Tate completion: $2m \mid L$ implies $U_j(t^L) \in \overline{\mathbb{Q}}[[t]]$. Therefore $s_j = U_j h_T$ at this cusp. Every product of $n$ unary factors lies in the same line $\mathcal{L} = (\pi^* H)^n$. By Lemma 3.1, the theta series are the expansions of genuine sections $\sigma_i$ of this line, in the common frame $h_T^n$.

For the composite map $Y \to X_0 \to X$ the degree is at most $24K^s$. Consequently

$$\deg \mathcal{L} \leq \frac{n}{2} 24K^s \deg \omega \leq \frac{nK^s I(L)}{2}.$$

At the chosen cusp, $w = \operatorname{ord}_y(q) \geq L$. Since $L \geq 24$ and $B_* = nK^s I(L)/48$, a linear combination vanishing through $q^{B_*}$ has section order at least

$$w(B_* + 1) \geq L(B_* + 1) > \frac{nK^s I(L)}{2} \geq \deg \mathcal{L}.$$

A nonzero global section of a line bundle on a smooth proper curve has zero divisor of degree equal to the degree of that line. The section must therefore be zero. Expanding in the common frame proves that the full formal linear combination is zero. This proves Theorem 2.1.

Notice that the numerical bound needs only $n, N, s$ and the finite group order $I(L)$. The kernel algorithm need not compute the root torsor, factor a Kummer extension, find its optimal degree, or enumerate cusp inertia cycles in order to use this sufficient bound.

# 7 Exact algorithms and practical stopping certificates

Each coefficient up to a given $B$ is computable by exact ellipsoid enumeration. A rational $LDL^t$ decomposition supplies positive rational diagonal entries and permits recursive completion of squares. At each stage integer interval endpoints are obtained with integer square roots and exact rational inequalities. The search is finite, enumerates every vector with $Q(x) \leq B$, and counts both signs without approximation. Exact Gaussian elimination over $\mathbb{Q}$ then computes $\ker M_B$, pivot columns and the coordinates of all columns. Base extension to $\mathbb{C}$ preserves the matrix rank and kernel dimension.

**Lemma 7.1** (Earlier certificates)**.** *For every $B \geq 0$, $\ker \Theta \subseteq \ker M_B$. If $M_B$ has full column rank, the complete theta kernel is zero. More generally, if a space $W$ of rigorously established full-series relations has dimension $h - \operatorname{rank} M_B$, then $W = \ker \Theta = \ker M_B$.*

*Proof.* Truncation of a full relation proves the inclusion. In the second case $W \subseteq \ker \Theta \subseteq \ker M_B$ and the two outer spaces have the same dimension. □

In particular, a singleton theta family has complete kernel zero already at $B = 0$, since its constant term is 1. This assertion is independent of modularity. An integral basis-change

certificate $P^tAP = A'$, $\det P = \pm 1$, proves a full-series relation by a norm-preserving bijection; merely matching a finite prefix proves no such relation.

The source companion implementation `exact-engine/theta_exact.py` returns full kernel and coordinate claims only when these stopping certificates justify them. Deficient truncations without sufficient full-series certificates remain explicitly unresolved. The companion `solve_universal.py` derives the bound internally and computes prefixes at $0, 1, 2, 4, \ldots$, capped at $B_\dagger$. It stops at an earlier certificate when available; otherwise it computes through $B_\dagger$ and applies Theorem 9.5. This is a terminating algorithm on every valid finite input and accepts no externally asserted bound. Its default command only inspects the bound; `--complete` executes the algorithm without an attempt budget. These companion commands describe the source algorithm; they are not required executables in this Lean delivery. The formal development proves the LDL enumeration and capped solver contracts. Symbolic bound inspection and actual completed coefficient enumeration are distinct. The theorem proves termination, not practical tractability of every input.

# 8 Two arithmetic examples

## 8.1 A complete order with three independent classes

In $(\mathbb{Q}^2, X^2 + 20Y^2)$, use bases

$$T_1 = I, \qquad T_2 = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}, \qquad T_3 = \operatorname{diag}(2, 1/2).$$

Their normalized forms are $x^2 + 20y^2$, $3x^2 + 2xy + 7y^2$, and $4x^2 + 5y^2$. The unnormalized restriction for $T_2$ is *three times* the second form; thus no false unscaled rational isometry is asserted. All normalized Gram matrices have Smith factors $(2, 40)$.

For a reduced form $ax^2 + bxy + cy^2$ with Gram determinant 80, $|b| \le a \le c$ implies $3a^2 \le 80$, hence $a \le 5$. Enumerating these finitely many possibilities and imposing the Smith condition leaves $(1, 0, 20), (3, \pm 2, 7), (4, 0, 5)$. The two signs are integrally equivalent. The minima $1, 3, 4$ distinguish the three classes, and all three have been realized in the fixed rational space. Their coefficient rows at $0, 1, 3$ form

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}, \qquad \det = 4.$$

The full kernel is zero. The delivered formal classification proves completeness, content normalization and rational-similarity assertions; its endpoint mapping is given below.

## 8.2 A full nonzero relation between distinct classes

Let $D_n = \{x \in \mathbb{Z}^n : \sum x_i \equiv 0 \pmod 2\}$ and $D_n^+ = D_n \cup ((1/2, \ldots, 1/2) + D_n)$ for $8 \mid n$. These are even unimodular lattices. Write

$$a = \sum_{r \in \mathbb{Z}} u^{4r^2}, \quad b = \sum_{r \in \mathbb{Z}} (-1)^r u^{4r^2}, \quad c = \sum_{r \in \mathbb{Z}} u^{(2r+1)^2}.$$

The four-dimensional Hadamard matrix divided by 2 bijects integral vectors of odd coordinate sum with one parity half of the half-integral vectors, preserving norm. Reflection in one coordinate exchanges the two half-integral parity halves. Consequently $a^4 = b^4 + c^4$ as

a formal identity. Parity selection gives the norm series of $D_n^+$ as $(a^n + b^n + c^n)/2$. The polynomial identity

$$\left(\frac{X^2+Y^2+Z^2}{2}\right)^2 = \frac{X^4+Y^4+Z^4}{2} \quad (X = Y + Z)$$

then proves $\theta_{D_8^+ \perp D_8^+} = \theta_{D_{16}^+}$ coefficientwise. Substitution $q = u^8$ is injective, so this is the original integral theta identity, not a finite check.

The root nonorthogonality graph of the orthogonal sum is disconnected, whereas that of $D_{16}^+$ is connected. The two lattices are not isometric. Since their determinants are both 1, any lattice similarity would have positive scale $s$ satisfying $s^{16} = 1$, hence would be an isometry. They are distinct rational-similarity classes in the same order. For these two supplied columns the full kernel is exactly $\mathbb{C}(1, -1)$. No unproved classification of the entire rank-sixteen order is required for this example.

# 9 Arithmetic normalization on torsion-level moduli

The following refinements give full formal expansions on specified arithmetic level structures. In particular they remove a common-unit ambiguity in a projective theta comparison. The comparison of the resulting half-weight lines is kept separate from these identities.

## 9.1 A theta fourth power from two-torsion coordinates

Write $P(Q) = \prod_{r\geq 1}(1 - Q^r)$. In the standard Tate Weierstrass model with invariant differential $\delta = dz/z$, the $x$-coordinate is

$$X(z; Q) = \sum_{r\in\mathbb{Z}} \frac{Q^r z}{(1 - Q^r z)^2} - 2\sum_{r\geq 1} \frac{rQ^r}{1 - Q^r}.$$

For negative indices the summands are rewritten using $v/(1 - v)^2 = v^{-1}/(1 - v^{-1})^2$; the resulting expressions are formal on each Tate annulus. To verify that these are actual algebraic coordinates, use the proper polygon model of [11, Section 2.5], after a finite parameter extension. On its annular charts the displayed inversion rewrites all negative-index terms with nonnegative parameter order. Away from the identity all denominators are units; at the identity only the indicated double pole remains. Thus $X$ is a section of $\mathcal{O}(2[0])$ on the formal model. If a further chart refinement introduces vertical denominators, a single parameter power clears them, since the cover is finite. Proper formal functions for this invertible sheaf algebraize that section, just as in Lemma 4.1.

Periodicity follows by shifting the summation index, and $X(z^{-1}) = X(z)$. The function $Y = \frac{1}{2} z\, \partial X/\partial z$ similarly algebraizes with a triple pole at the identity and is odd under inversion. The degree-two function $X$ gives the quotient by inversion, so $Y^2$ is a cubic in $X$. Its leading coefficient is one by the Laurent terms $X(1 + s) = s^{-2} + O(s^{-1})$ and $Y(1 + s) = -s^{-3} + O(s^{-2})$. Its three roots are the $X$-coordinates of the nonzero two-torsion points. This proves $Y^2 = \prod_j (X - e_j)$ and $dX/(2Y) = dz/z$ on the actual algebraic Tate curve, rather than assuming that an arbitrary periodic formal expression is rational.

**Lemma 9.1** (Exact basic theta normalization)**.** *For $\vartheta(q) = \sum_{r\in\mathbb{Z}} q^{r^2}$ one has the full formal identity*

$$\vartheta(q)^4 = 4\big(X(q; q^2) - X(-1; q^2)\big).$$

*Proof.* First the multiplicative addition formula is

$$X(z;Q) - X(w;Q) = \frac{-wP(Q)^4\gamma_Q(z/w)\gamma_Q(zw)}{\gamma_Q(z)^2\gamma_Q(w)^2}. \tag{2}$$

For generic $w$, both sides are periodic in $z$, have zero divisor $[w] + [w^{-1}]$ and pole divisor $2[1]$. Their quotient is constant by Lemma 4.1, applied to a proper Tate model after a finite extension splitting the points. The local horizontal factors of each $\gamma$ are its indicated simple zeros; the remaining products are units, as in the proof of Proposition 4.2. The same comparison may be made with $w$ indeterminate and then specialized as a rational identity. At $z = 1 + s$, use $\gamma_Q(1 + s) = -sP(Q)^2 + O(s^2)$ and $\gamma_Q(w^{-1}) = -w^{-1}\gamma_Q(w)$. The right side has leading term $s^{-2}$, equal to that of $X(1+s) - X(w)$, so the constant is one. Specialization to nonzero two-torsion is valid; the two simple zeros then coalesce to a double zero.

Set $Q = q^2, z = q, w = -1$, and abbreviate

$$A = \prod_{r\geq 1}(1 - q^{2r-1}), \quad B = \prod_{r\geq 1}(1 + q^{2r-1}), \quad C = \prod_{r\geq 1}(1 + q^{2r}).$$

Then $\gamma_{q^2}(q) = A^2$, $\gamma_{q^2}(-q) = B^2$ and $\gamma_{q^2}(-1) = 2C^2$. The Euler identity

$$\prod_{r\geq 1}(1 - T^{2r-1}) \prod_{r\geq 1}(1 + T^r) = 1$$

gives $ABC = 1$. Thus (2) becomes

$$4(X(q;q^2) - X(-1;q^2)) = P(q^2)^4B^4/(A^4C^4) = P(q^2)^4B^8.$$

The formal triple product already proved gives $\vartheta(q) = P(q^2)B^2$, proving the identity in all degrees. □

**Proposition 9.2** (A square root supplied by four-torsion)**.** *On a characteristic-zero elliptic curve with labelled nonzero two-torsion $P_j = (e_j, 0)$, write*

$$Y^2 = (X - e_1)(X - e_2)(X - e_3), \qquad \delta = dX/(2Y).$$

*Choose a four-torsion section $R$ with $2R = P_2$. Then*

$$g = \frac{(X(R) - e_1)^2 - (e_1 - e_2)(e_1 - e_3)}{2Y(R)}, \qquad \alpha = 2g\delta$$

*defines a weight-one section on this arithmetic level structure, and $\alpha^2 = 4(e_2 - e_1)\delta^2$. At the Tate marking $Q = q^2$, $P_1 : z = -1$, $P_2 : z = q$, $R : z = q^{1/2}$, its expansion is*

$$\alpha = \vartheta(q)^2\delta.$$

*Proof.* Put $u = X - e_1$, $a = e_1 - e_2$, $b = e_1 - e_3$. The tangent-and-chord law on $Y^2 = u^3 + (a + b)u^2 + abu$ gives

$$X(2R) - e_1 = \frac{(u(R)^2 - ab)^2}{4Y(R)^2}.$$

Since $2R = P_2$, this proves $g^2 = e_2 - e_1$. The denominator is nonzero because $R$ has exact order four. Under a Weierstrass coordinate change $g$ has weight one; hence $2g\delta$ is intrinsic. At the specified Tate marking, $X(R) = q^{1/2} + O(q)$, $Y(R) = \frac{1}{2}q^{1/2} + O(q)$, $e_1 = -1/4 + O(q^2)$ and $e_2 = 2q + O(q^2)$, so $g = 1/2 + O(q^{1/2})$. Lemma 9.1 identifies its square, and the leading sign selects exactly $\vartheta(q)^2$. □

At a full-level cusp, the nonzero two-torsion representatives are $-1, Q^{1/2}, -Q^{1/2}$, in some order. The difference of the last two $X$-coordinates has $Q$-order $1/2$; either difference involving the first has order zero. Thus $\alpha$ extends at every cusp, with $Q$-order zero or $1/4$, and it is nowhere zero on the open curve. On a full level $M$ divisible by eight, $Q = t^M$, so $\operatorname{div}(\alpha) = 2D$ for an integral effective cusp divisor $D$. Consequently $H = \mathcal{O}(D)$, $s = 1_D$ and $s^2 \mapsto \alpha$ give a specific algebraic square root $H^2 \simeq \omega$. Its prescribed Tate frame is $h = \vartheta(q)^{-1}s$, with $h^2 = \delta$ and $s = \vartheta(q)h$. Here $q = Q^{1/2}$ on the specified cusp.

For $\vartheta(q^d)$, pull this construction back along the cyclic isogeny $E_q \to E_{q^{2d}}$, $z \mapsto z^{2d}$. The target Hodge section must be multiplied by $(2d)^{-1}$, since the target differential pulls back to $2d\, d\log z$. A full source level divisible by $2dM$ supplies the required target torsion lifts. This produces a normalized half-weight section for each $d$, but does not yet assert that all such half-weight lines have been identified on one torsion level.

## 9.2 The exact finite theta-multiplier character

The remaining fourth-root phase of Proposition 9.2 can also be computed arithmetically. This identifies the basic multiplier, without asserting compatibility for different isogenies.

**Proposition 9.3** (A full-four-level character)**.** *Let $b = (R_1, R_2)$ be an ordered full-four basis, $P_i = 2R_i$, $P_3 = P_1 + P_2$, and $\zeta = e_4(R_1, R_2)$. For the right action on bases of the subgroup*

$$G = \{g \in \mathrm{SL}_2(\mathbb{Z}/4) : g \bmod 2 \text{ preserves } xy\},$$

*the section $\alpha$ of Proposition 9.2 satisfies $\alpha(bg) = \chi(g)\alpha(b)$, where*

$$\chi(S) = \zeta^{-1}, \qquad \chi(T^2) = 1, \qquad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

*This is exactly the genus-one even discriminant character $\lambda$ of [16, Section 2.2], with these Weil-pairing and ordered-basis conventions.*

*Proof.* Translate $e_1$ to zero and write $Y^2 = X(X + a)(X + b)$, $v = X(R_1)$, $u = X(R_2)$ and $y_i = Y(R_i)$. The duplication conditions imply

$$v^2 = ab, \qquad u^2 + 2au + ab = 0, \qquad y_1^2 = v(v + a)(v + b), \quad y_2^2 = u(u + a)(u + b).$$

Our root is $g_{12} = -a(u + b)/y_2$. Under $S$ the ordered basis becomes $(R_2, -R_1)$, and direct substitution into its defining formula gives

$$g_{12}(bS) = -a(v + b)/y_1, \qquad \chi(S) = \frac{(v + b)y_2}{(u + b)y_1}.$$

To fix this fourth-root phase, use normalized Miller functions. If $\ell_i$ is the tangent linear function at $R_i$, then

$$\operatorname{div}(\ell_i) = 2[R_i] + [P_i] - 3[0], \qquad f_i = \ell_i^2/(X - e_i), \quad \operatorname{div}(f_i) = 4[R_i] - 4[0].$$

At the identity, in parameter $-X/Y$, $f_i$ has leading coefficient one. Algebraic Weil reciprocity, applied after moving the supports and then taking the limit at the identity, gives $e_4(R_1, R_2) = f_1(R_2)/f_2(R_1)$; the general normalization sign is $(-1)^4 = 1$. The tangents pass through $P_i$, so

$$\ell_1(R_2) = y_2 - y_1 u/v, \qquad \ell_2(R_1) = y_1 - y_2(v + a)/(u + a).$$

For $t = y_1/y_2$, substitution yields

$$e_4(R_1, R_2) = \frac{(v + a)(u + a)^2(v - tu)^2}{uv^2((u + a)t - v - a)^2} = \frac{(u + b)t}{v + b} = \chi(S)^{-1}.$$

For explicit verification, after clearing denominators the middle identity reads

$$(v+a)(v+b)(u+a)^2(v-tu)^2 - uv^2(u+b)t((u+a)t-v-a)^2 = 0.$$

It follows by substituting $v^2 = ab$, $u^2 = -2au - ab$ and $t^2u(u+a)(u+b) = v(v+a)(v+b)$. The denominators do not vanish on the full-four-level moduli problem: the distinct halves have order four and the tangent divisors avoid the evaluation points. This proves $\chi(S) = \zeta^{-1}$.

Adding $P_1$ sends $u$ to $ab/u$ and $y_2$ to $-aby_2/u^2$; the formula $(u^2 - ab)/(2y_2)$ is unchanged. Hence $\chi(T^2) = 1$. The mod-two kernel in $G$ consists of $I + 2A$ with $\mathrm{tr}(A) = 0$ and has order eight. It is generated by $S^2 = -I$, $T^2$ and $ST^2S^{-1}$; the remaining mod-two image is $\{I, S\}$. Thus $|G| = 16$ and $S, T^2$ generate it. The computed ratios, independent of the curve, multiply under successive changes of marking, and therefore define the claimed full character. The defining values of $\lambda$ in the cited source are $\lambda(S) = -1$, $\lambda(T^2) = 0$ in additive $\mathbb{Z}/4$ notation. They agree with the values above under $j \mapsto \zeta^j$. □

This also fixes the descent direction. The theta multiplier line $M_\Theta$ uses the torsor of isomorphisms from $E[4]$ to the standard module, whereas our ordered basis is an isomorphism in the opposite direction. Its level frame consequently satisfies $m_{bg} = \lambda(g)^{-1}m_b$. Therefore

$$\alpha(b) \otimes m_b \in \omega \otimes M_\Theta$$

is independent of the full-four marking and descends to the moduli problem with the even characteristic $P_3$. For the rigidified line $\Theta = \mathcal{O}(P_3)$, let $J = \pi_*\Theta$. Its identity-evaluation section $e \in J^{-1}$ is nowhere zero on the open moduli problem, because $P_3$ is disjoint from the identity. Thus the rule

$$J^{-2} \xrightarrow{\sim} \omega \otimes M_\Theta, \qquad e^2 \longmapsto \alpha(b) \otimes m_b$$

is an explicit algebraic key-formula isomorphism for this genus-one even characteristic. Its full Tate normalization is fixed by Proposition 9.2; no claim that all units on an open modular curve are constant has been used.

### 9.3 Projective theta values without an unknown formal unit

**Proposition 9.4** (Exact projective unary ratios)**.** *For every positive even $m$ there are algebraic rational functions $R_{m,\nu}$ on an arithmetic full $2m$-level curve, with a specified compatible theta marking, such that their complete Tate expansions are*

$$R_{m,\nu}(q) = \frac{U_{m,\nu}(q)}{U_{m,0}(q)}, \qquad U_{m,0}(q) = \vartheta(q^{m/2}).$$

*Proof.* Use the rigidified totally symmetric degree-$m$ line $\mathcal{L}_m = \mathcal{O}_E(m[0]) \otimes \omega_E^m$. The algebraic full-level domination of symmetric theta structures in [15, Remark 2.6.5] supplies the structure on full level $2m$. The algebraic Schrödinger comparison [15, Theorems 4.1.1 and 4.1.3] identifies $\pi_*\mathcal{L}_m$ with the standard $m$-dimensional representation tensored with one line. Evaluation at the identity therefore has components $s_{m,\nu}$ in a common line. Their quotients are rational functions, independent of the trivialization of that line.

Here is the complete formal normalization, including relative phases. On the Tate multiplicative cover the section equation is

$$f(qz) = q^{-m/2}z^{-m}f(z).$$

Indeed $z^{-m/2}\gamma_q(z)^m$ has this multiplier and divisor $m[0]$; the rigidification changes only the common base frame. On full $2m$ level choose the symmetric theta marking with operators

$$(Af)(z) = f(\zeta_m z), \qquad (Bf)(z) = q^{1/(2m)}zf(q^{1/m}z).$$

They satisfy $A^m = B^m = 1$, $AB = \zeta_m BA$, and inversion conjugates each to its inverse. The extra root $q^{1/(2m)}$ belongs to the specified full level, not to an independently chosen completion extension. To justify this marking globally, the theta structure supplied by [15, Remark 2.6.5] induces the prescribed full $m$-level structure. Its difference from the displayed symmetric lifts, which have the same underlying level, belongs to $K(m)[2]$ by [15, Lemma 2.5.3]. On full level this is a constant finite group. On the chosen connected Tate branch the difference is therefore one constant element; composing with that same global theta-group automorphism aligns the lifts. This correction changes only finite signs and adds neither a parameter-dependent factor nor a new cover.

The exact eigenbasis is

$$\vartheta_{m,\nu}(z) = \sum_{r \equiv \nu\,(m)} q^{r^2/(2m)} z^r.$$

The positive quadratic exponent makes each Laurent sum convergent in the formal sense on every annulus: substituting $z = t^j x$ leaves parameter orders tending quadratically to infinity in both directions. The prescribed automorphy glues these as formal sections of the actual line $\mathcal{L}_m$. A finite cover bounds any vertical denominators; proper formal functions for that line then algebraize the sections. The generic section space has dimension $m$ by Riemann–Roch on the elliptic curve. Thus these are actual section vectors, not only solutions in an unspecified Laurent space. Its $A$-eigenvalue is $\zeta_m^\nu$, and $B\vartheta_{m,\nu} = \vartheta_{m,\nu+1}$ without an extra factor. For any section $\sum a_r z^r$, its equation implies $a_{r+m} = q^{r+m/2} a_r$. Thus each $A$-eigenspace is one-dimensional. An intertwiner sends the standard basis to $b_\nu \vartheta_{m,\nu}$; the $B$ action forces all $b_\nu$ to be one and the same nonzero scalar. Evaluating at $z = 1$ and dividing by the zeroth component cancels this scalar and the rigidification frame. The denominator is not identically zero, since $U_{m,0}$ has constant term one. This proves the asserted rational function and every coefficient of its expansion. No half-Hodge comparison has been used. □

These two constructions reduce an arithmetic half-weight comparison to basic theta lines: formally, $U_{m,\nu} = R_{m,\nu}\vartheta(q^{m/2})$. Once a candidate section has been put in a specified half-Hodge line, its regularity need not be guessed from its rational-function factors. Its $K$th power equals $F_{m,\nu}$ by the injection of the function field into the designated Tate completion, since both complete expansions agree. Thus its order at every point is nonnegative by the already proved regularity of $F_{m,\nu}$.

### 9.4 A common arithmetic line and a smaller coefficient bound

The torsion-level constructions above give a much smaller common cover than adjoining every $K$th root separately. This observation closes the common-line construction for the arithmetic coefficient theorem; it does not identify the resulting finite cover as a congruence cover.

**Theorem 9.5** (A smaller arithmetic finite-coefficient bound)**.** *Let $r$ be the number of distinct positive even integers $m_{ij}$ in the orthogonal decompositions, and retain $N, L$ as above. Then one may replace $B_*$ in Theorem 2.1 by*

$$B_\dagger = \left\lfloor \frac{n\, 2^{r-1} I(L)}{48L} \right\rfloor.$$

*The proof uses the arithmetic level constructions of this section and none of the complex comparison in Section 10.*

*Proof.* For each distinct $m$, the full source level $L = 24N^2$ supplies the cyclic $m$-isogeny $\phi_m : E \to E_m$ and a full-four marking on its target: $4m \mid L$. At the distinguished cusp the target period is $q^m$. Pull back the section $\alpha$ of Proposition 9.2 and divide by $m$, using the

Hodge pullback isomorphism. Call the resulting source weight-one section $a_m$. Its exact expansion is

$$a_m = \vartheta(q^{m/2})^2\delta = U_{m,0}(q)^2\delta.$$

This normalization fixes the full series and its sign simultaneously.

At an arbitrary source cusp write $q_x = t^L$. If the toric part of the cyclic kernel is $\mu_e$, the residual component part has order $b = m/e$. The target period satisfies $Q_x^b = \zeta q_x^e$, so

$$\operatorname{ord}_t(Q_x) = Le^2/m.$$

The Hodge map there is multiplication by $e$; the factor $e/m$ is a nonzero constant and introduces no zero or pole. The earlier four-torsion calculation shows that $a_m$ has order either zero or $Le^2/(4m)$. Since $8m \mid L$, both possibilities are even integers. It has no interior zero, since the two distinct two-torsion coordinates never coincide. Hence $\operatorname{div}(a_m) = 2D_m$ for an effective integral cusp divisor $D_m$ on $X$. Set

$$H_m = \mathcal{O}_X(D_m), \qquad e_m = 1_{D_m}, \qquad H_m^2 \xrightarrow{\sim} \omega, \quad e_m^2 \longmapsto a_m.$$

The equality of divisors proves that this is an everywhere invertible line-bundle map, including the boundary.

Fix one value $m_0$ and write $H_0 = H_{m_0}$. For each of the other $r-1$ values, $T_m = H_m \otimes H_0^{-1}$ has a specified trivialization $T_m^2 \simeq \mathcal{O}_X$. Its unit-frame torsor is therefore finite etale of degree two over the entire compact curve. Form their fiber product and choose the component with the following common Tate branch:

$$t_m = \frac{U_{m_0,0}(q)}{U_{m,0}(q)}\,\frac{e_m}{e_{m_0}}, \qquad t_m^2 = 1.$$

All these series have constant term one, and both displayed sections are nonvanishing at the distinguished cusp. Thus the branch already exists over its completed local ring. This gives a finite etale map $p : Y \to X$ of degree at most $2^{r-1}$, unramified at that cusp. For $r = 1$ take $Y = X$.

Dividing $p^*e_m$ by the tautological frame $t_m$ gives a section $s_{m,0}$ of the common line $p^*H_0$. With $h_0 = U_{m_0,0}^{-1}e_{m_0}$, one has $h_0^2 = \delta$ and $s_{m,0} = U_{m,0}h_0$ at the selected cusp. Define the rational section

$$s_{m,\nu} = p^*R_{m,\nu}\, s_{m,0}.$$

Proposition 9.4 gives its exact expansion $U_{m,\nu}h_0$. Under $(p^*H_0)^K \simeq p^*\omega^{K/2}$ its $K$th power consequently equals $p^*F_{m,\nu}$: their full expansions agree, and the function field injects into the cusp completion. Since the latter is regular everywhere, $K \operatorname{ord}_y(s_{m,\nu}) \geq 0$ at every point $y$, so $s_{m,\nu}$ is regular. This also handles possible poles in the rational factors $R$.

The original coset expressions now give all $\theta_i$ as sections of one line $\mathcal{L} = (p^*H_0)^n$, with a single common Tate frame $h_0^n$. Using $24 \deg \omega \leq I(L)$ yields

$$\deg \mathcal{L} = \frac{n}{2}\deg(p)\deg\omega \leq \frac{n\,2^{r-1}I(L)}{48}.$$

The selected cusp still has $\operatorname{ord}(q) = L$. A nonzero section cannot vanish to order $L(B_\dagger + 1)$, since this exceeds the displayed degree bound by the definition of the floor. Thus vanishing of the coefficients through $B_\dagger$ forces the complete section, and therefore its formal theta series, to vanish. Base change to $\mathbb{C}$ gives the same statement for all complex linear combinations. □

The new bound replaces the factor $K^s$ by $2^{r-1}/L$. Its numerical advantage is independent of the early stopping certificates of Lemma 7.1: those certificates can still finish long before even $B_\dagger$. Neither the compact etale torsors used here nor their characters have been asserted to descend to a principal congruence level merely because they are finite.

### 9.5 Arithmetic modularity and the image basis

**Corollary 9.6** (Modularity on a standard finite metaplectic level)**.** *The sections constructed in Theorem 9.5 are standard arithmetic modular forms of weight $n/2$ at a common finite metaplectic level. Their complete Tate expansions are the input $\theta_i$. Pivot columns of $M_{B_\dagger}$ give a linearly independent modular-form basis of their image, and exact rational elimination gives the coordinates of every $\theta_i$ in that basis.*

*Proof.* Let $X^\circ$ be the open full-level curve and $\mathcal{X}_{1/2} = \sqrt[2]{\omega/X^\circ}$ the algebraic metaplectic stack of [15, Definition 3.1.1]. Its object over a test scheme $T$ is a level-marked elliptic curve, a line $R$, and a specified isomorphism $R^2 \simeq \omega_T$. Denote its tautological line by $\lambda$. The actual isomorphism $H^2 \simeq p^*\omega$, with $H = p^*H_0$, defines

$$j : Y^\circ \longrightarrow \mathcal{X}_{1/2}, \qquad j^*\lambda = H.$$

This is a genuine finite metaplectic level: for a test object $(T, R, R^2 \simeq \omega_T)$, the fiber product $Y^\circ \times_{\mathcal{X}_{1/2}} T$ is the torsor

$$\mathrm{Isom}_{\mathrm{sq}}(R, H) \longrightarrow Y^\circ \times_{X^\circ} T$$

of isomorphisms compatible with the two specified square maps. It is a representable finite etale $\mu_2$-torsor. Since $p$ is finite etale, $j$ is representable finite etale, of degree $2 \deg p$. Thus both the level and its weight line arise from the pre-existing metaplectic moduli functor.

The same square isomorphism holds on the entire compact curve $Y$. The sections of $H^n$ are regular at every cusp by the proof of Theorem 9.5, and their expansions in the common nonvanishing frame $h_0^n$ are exactly $\theta_i(q)$. Hence these are modular forms of weight $n/2$ in the arithmetic definition, with holomorphy at the boundary, not merely sections on the open curve. Their linear relations coincide with their formal-series relations by injectivity at the chosen cusp. Theorem 9.5 and rational elimination now give the asserted basis and coordinates. □

## 10 Scope of the arithmetic modular-form conclusion

The modular-form assertion just proved uses the existing algebraic metaplectic definition and an explicitly constructed finite level. Its proof contains no appeal to complex realization of a cover, upper-half-plane simple connectivity, analytic theta transformation, or comparison of an analytic Fourier frame with a formal Tate frame. Every required frame and every coefficient was instead constructed inside one algebraic formal completion.

The conventional description of such objects by functions on the upper half-plane belongs to the general comparison between arithmetic and analytic definitions of modular forms. That comparison is not an input to our coefficient theorem, our construction of the modular forms, or the identification of their prescribed formal expansions. In particular we do not present the comparison theorem itself as an arithmetic theorem proved here. An analytic reformulation would use that separate general dictionary; none is needed for the original formal-series task formulated with arithmetic modular forms.

The finite frame level $Y$ may be noncongruence. Its finiteness and etaleness do not imply domination by a principal congruence level, and we have made no such assertion. Likewise, the specific Shimura half-form lines in [15, Definitions 3.4.1 and 3.4.4] include additional theta-multiplier data. We do not claim that the common level has a prescribed Shimura multiplier. These distinctions do not restrict the input family or the determination of all its formal relations: the input question specifies neither a congruence group nor a particular multiplier.

Full-series proofs are distinguished from finite experiments throughout. The bibliography identifies the external mathematical background. The delivered Lean source closure proves the frozen endpoints without adding bibliographic claims as axioms. Historical proof records and the original source are immutable provenance; a historical acceptance label alone is not current integration acceptance.

# 11 Formalization and verification

## 11.1 The explicit binary lattice

**Theorem 11.1** (Complete binary coset identity)**.** *Let*

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 12 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}.$$

*Then $A$ is positive definite and even integral,* $\det B = 2$*, and* $B^t AB = \operatorname{diag}(2, 46)$*. The two classes in $B^{-1}\mathbb{Z}^2/\mathbb{Z}^2$ have representatives $(0, 0)$ and $(1/2, 1/2)$, with unary residues $(0, 0)$ and $(1, 23)$. For every nonzero integer $D \geq 0$ divisible by 92, put*

$$V_{m,w,\nu}(q) = \sum_{r \equiv \nu \pmod m} q^{wr^2}.$$

*Then the integral theta series satisfies*

$$\theta_A(q^D) = V_{2,D/4,0}(q)V_{46,D/92,0}(q) + V_{2,D/4,1}(q)V_{46,D/92,23}(q).$$

*The unexpanded equality is the sum of the two integral coset series.*

*Proof.* The norm is $x^2 + xy + 6y^2$, positive away from zero by completion of squares. Direct multiplication gives the diagonalization and determinant. The displayed representatives map under $B$ to $(0, 0)$ and $(0, 1)$, so belong to the preimage lattice. Their difference is not integral, since its first coordinate is $-1/2$. The determinant-index theorem makes the quotient cardinality two, hence these distinct classes exhaust it. Multiplication by the norms gives the asserted residues. Apply the general orthogonal coset factorization with these representatives. The divisibilities $4 \mid D$ and $92 \mid D$ make each unary exponent integral, proving the displayed complete series identity. Power-series expansion by a positive integer is injective, which explains the fractional-exponent notation of the original example without replacing it by a finite computation. □

## 11.2 One modular family for all certified conservative prefixes

**Theorem 11.2** (Conservative image assembly)**.** *For a nonempty valid family of rank $n$, there is exactly the conservative cutoff $B = B_*$ described above when $n > 0$, and $B = 0$ when $n = 0$, such that at every $B' \geq B$ rational elimination of $M_{B'}$ gives a basis of the complete complex theta kernel, independent pivot theta series spanning the entire image, and unique rational coordinates for each input series. For $n > 0$, one compact modular family over $\mathbb{Q}(\zeta_L)$ realizes these same image and coordinate conclusions at every $B' \geq B$, using its structural scalar map and its fixed global sections.*

*Proof.* Take $B$ and the complete kernel equality from the conservative construction. For a given $B' \geq B$, set $W$ equal to the actual full theta kernel. The kernel equality and rational matrix nullity give the exact dimension certificate for rational elimination. Its null vectors span the full complex kernel, and its pivot columns are an independent spanning family.

Elimination coordinates reconstruct every complete series; independence proves uniqueness. No full-rank assumption is needed. In rank zero every theta series is 1, and the full relation map is the constant series with value $\sum_i c_i$.

In positive rank use the constructed cyclotomic compact family, denoted $G$. Its completed-expansion theorem identifies relations among $G$'s sections with relations among their full theta series, for all coefficient vectors over the base field. The certified prefix theorem extends the rational elimination conclusions to that field. Inserting pivot coefficients into the full coefficient vector transfers independence and existence of the pivot expansion through the relation equivalence. Applying the same equivalence to two pivot expansions proves uniqueness. A single-column vector transfers the explicit elimination coordinates. All these steps keep $G$, its sections and its structural scalar map fixed. They apply separately to every certified $B'$, so no numerical comparison with the sharper cutoff is required. □

### 11.3 Formal objects and complete source coverage

The development represents theta coefficients by cardinalities of actual integer lattice shells, uses exact rational Gram matrices and elimination, and interprets unary expansions as formal power series. Positive definiteness supplies finite shells. The algebraic construction uses the actual proper model, normal completed chart domains and fraction maps. The arbitrary-cover constant theorem retains its smooth geometrically connected generic fiber, properness, flatness and height-one no-pole hypotheses; it is not restricted to a chosen cyclic chart model.

The norm theorem constructs the Kato rational function, its Cartier divisor, all positive coprime multiplication norms, base-change compatibility and complete Tate product. The discriminant theorem identifies the canonical conormal tensor with five times the five-Kato initial term and its entire $Q^2\prod(1-Q^r)^{48}$ expansion. The generated field for simultaneous roots uses the specified ambient completed embedding; its field structures, degree, finite constants and actual regular sections are retained.

The sharp construction distinguishes the number $r$ of norms from the number $s$ of unary pairs. The cyclic Hodge factor is $e/m$ at a general cusp, and the target-period order is $Le^2/m$. The four-torsion orders are zero or $Le^2/(4m)$; $8m \mid L$ makes them even. The $r = 1$ case uses the identity cover. The multiplier character uses the stated right action and Weil-pairing convention; the inverse frame action ensures descent. Identity evaluation is proved nonvanishing, and the key isomorphism is normalized in its actual frame. Projective theta components share one invariant evaluation line and one marked Cartesian disk, so their scalar cancels in the complete quotient. The round-266 global provider and round-267 coherent-sections recovery are retained without renaming accepted declarations.

The finite metaplectic level is a map to the pre-existing root-stack moduli problem. Its test fibers are actual representable finite etale square-isomorphism torsors. The same compact square line carries the theta sections at the boundary. The cyclotomic family is finite dimensional over $\mathbb{Q}$ and all scalar actions arise from its specified structural map. These geometric providers are proved constructions, not arbitrary proposition fields standing in for geometry.

The determinant-80 proof includes the finite reduced-class enumeration, Smith invariants, rational embeddings and the factor 3 in the second embedding. The rank-16 proof includes the general $D_n^+$ even-unimodular and duality results, the Hadamard bijection, injective substitution and root-graph nonsimilarity. The computational development proves both signs of enumeration, exact shifted interval endpoints, rational Schur completion, LDL correctness and the terminating capped schedule. A deficient prefix without enough established full-series relations remains unresolved.

The accompanying `FormalizationMapping.md` lists all thirteen source requirements, supporting assertions, and all 103 exact frozen obligations. The two new declarations

are in namespace `ArithmeticThetaFullSourceR0268`, with names `binary_orthogonal_example` and `conservative_image_consumer`. The other 101 endpoints retain their accepted names and proofs. `Lean/FullSourceIntegrationR0268.lean` imports the endpoint closure; `Lean/GoalRoot.lean` imports all 1,198 supplied modules together. Module counts include supporting definitions and lemmas, not just endpoints.

### 11.4 Reproducible verification and evidence

The compiler is Lean 4.33.1. Mathlib and auxiliary packages are pinned by the checkout's manifest; the exact Mathlib revision is recorded in `Reproduce.md`. Build objects are outside the delivered source tree. The configured audit cache may supply unchanged hash-verified objects, while new modules and invalidated dependents are compiled with the same toolchain. Cache reuse does not replace current source, import, exact-type or axiom checks.

The independent auditor checks the 103 frozen propositions in the combined environment, rejects unresolved metavariables and proof placeholders, and checks the standard-only axiom closure. It compiles and verifies this PDF and records current artifact and input bindings in `AuditReport.json`, `ProofStatus.json` and `DeliveryManifest.json`. Those generated reports determine acceptance; this prose does not issue a verification receipt. The parent controller separately checks acceptance and whole-source mathematical coverage.

## 12 Limitations

The arithmetic modular forms here need not have principal-congruence level or a prescribed Shimura multiplier. The external analytic dictionary is not proved or used. Completeness of a supplied order's representative list remains an input hypothesis, except for the determinant-80 classification proved explicitly. The rank-16 example does not classify its entire order. Termination of exact enumeration does not give a practical complexity bound. The original companion Python commands are described for provenance; the formal deliverable and reproduction procedure are the Lean sources and audit. Compiler verification checks formal propositions and their dependencies; natural-language source correspondence also requires the independent coverage review. No historical accepted receipt substitutes for the current root audit.

## 13 Conclusion

Orthogonal cosets, normalized algebraic theta units and actual finite covers reduce complete lattice-theta relations to certified finite rational linear algebra. The two independent geometric constructions retain both explicit bounds, common full Tate expansions and regularity at every cusp. The final assembly covers the binary example, deficient rank, rank zero and unique image coordinates, with a fixed arithmetic modular family across all certified conservative prefixes.

Together, these constructions provide an explicit arithmetic route from finite lattice data to the complete theta relation kernel and image coordinates. The Lean development verifies the connecting steps, from lattice decompositions and normalized Tate expansions to geometric coefficient bounds and exact linear algebra. The resulting framework combines general finite-family input, explicit bounds and a documented formal proof chain within a single construction.

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